DAG and Simulation

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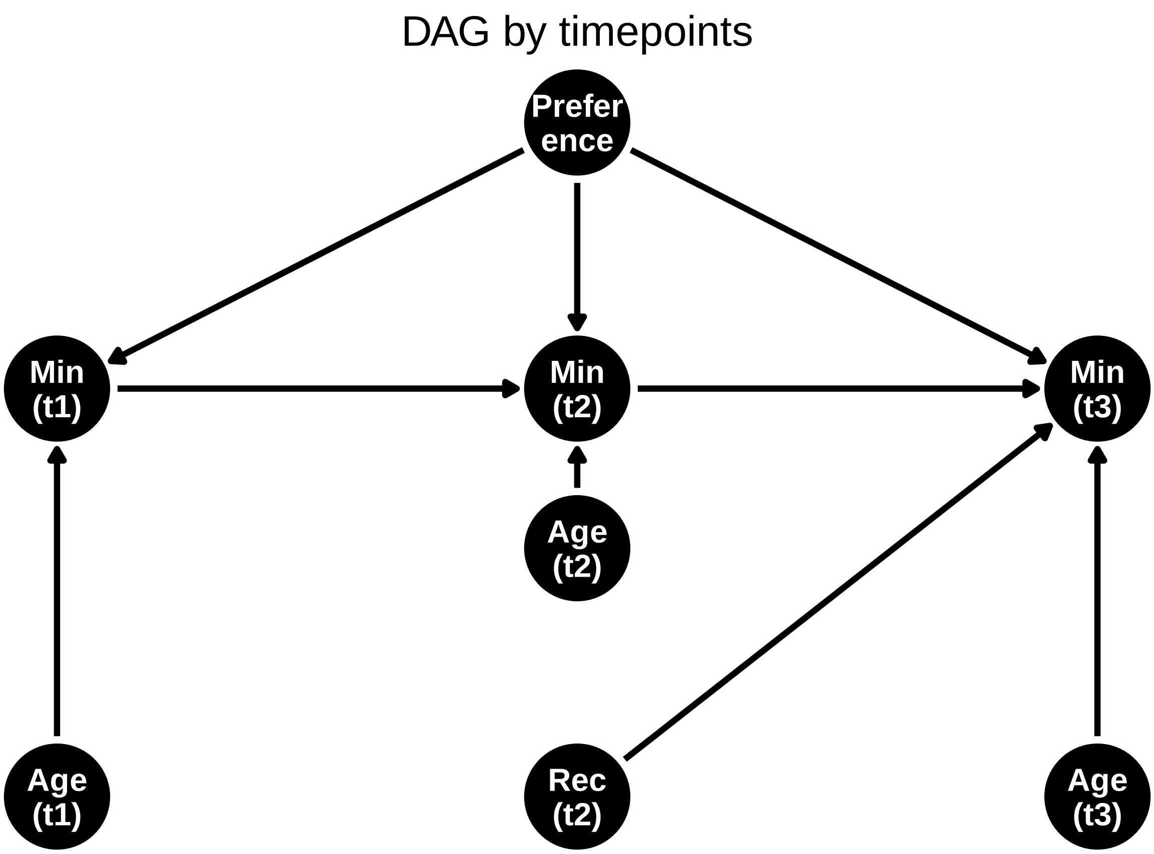
DAG

```
library(dagitty)
library(ggplot2)
library(ggdag)
```

```
##
## Attaching package: 'ggdag'

## The following object is masked from 'package:stats':
##
##     filter
```

```
dag_wb <- dagitty("
dag {
  Preference
  Min_viewed_t1
  Min_t2
  Min_t3
  Age_t1
  Age_t2
  Age_t3
  Rec_t2

  Preference -> Min_viewed_t1
  Preference -> Min_t2
  Preference -> Min_t3

  Min_viewed_t1 -> Min_t2
  Min_t2 -> Min_t3

  Age_t1 -> Min_viewed_t1
  Age_t2 -> Min_t2
  Age_t3 -> Min_t3

  Rec_t2 -> Min_t3
}
")

coordinates(dag_wb) <- list(
  x = c(
    Preference      = 0,
    Min_viewed_t1   = -3.2,
```

```r
    Min_t2          = 0,
    Min_t3          = 3.2,
    Age_t1          = -3.2,
    Age_t2          = 0,
    Age_t3          = 3.2,
    Rec_t2          = 0
  ),
  y = c(
    Preference      = 2.5,
    Min_viewed_t1   = 1,
    Min_t2          = 1,
    Min_t3          = 1,
    Age_t1          = -1.3,
    Age_t2          = 0.1,
    Age_t3          = -1.3,
    Rec_t2          = -1.3
  )
)

label_map <- c(
  Preference    = "Prefer\nence",
  Min_viewed_t1 = "Min\n(t1)",
  Min_t2        = "Min\n(t2)",
  Min_t3        = "Min\n(t3)",
  Rec_t2        = "Rec\n(t2)",
  Age_t1        = "Age\n(t1)",
  Age_t2        = "Age\n(t2)",
  Age_t3        = "Age\n(t3)"
)

p <- ggdag(dag_wb, text = FALSE) +
  geom_dag_edges(edge_width = 1.1) +
  geom_dag_point(size = 18) +
  geom_dag_text(aes(label = label_map[name]),
                color = "white",
                size = 4.2,
                fontface = "bold",
                lineheight = 0.9) +
  theme_void() +
  labs(title = "DAG by timepoints") +
  theme(plot.title = element_text(hjust = 0.5, size = 16))

print(p)
```



Factorization of the joint distribution

$$P(O) = P(\text{Preference})\, P(\text{Age}_{t1})\, P(\text{Min\_viewed}_{t1} \mid \text{Preference}, \text{Age}_{t1})\, P(\text{Age}_{t2})\, P(\text{Min}_{t2} \mid \text{Preference}, \text{Age}_{t2}, \text{Min\_viewed}_{t1})$$
$$\times\, P(\text{Rec}_{t2})\, P(\text{Age}_{t3})\, P(\text{Min}_{t3} \mid \text{Preference}, \text{Age}_{t3}, \text{Min}_{t2}, \text{Rec}_{t2}).$$

Simulation

```r
set.seed(1)
n <- 5000

Preference <- rnorm(n, 0, 1)

Age_t1 <- rnorm(n, 30, 8)
Age_t2 <- Age_t1 + 1
Age_t3 <- Age_t2 + 1

Min_viewed_t1 <- 20 + 4.5*Preference + 0.25*Age_t1 + rnorm(n, 0, 5)

Min_t2 <- 10 + 4.5*Preference + 0.20*Age_t2 + 0.50*Min_viewed_t1 + rnorm(n, 0, 5)

Rec_t2 <- rbinom(n, 1, 0.5)

Min_t3 <- 8 + 4.5*Preference + 0.18*Age_t3 + 0.45*Min_t2 + 6*Rec_t2 + rnorm(n, 0, 5)

data_3t_new <- data.frame(
```

```r
  Preference,
  Age_t1, Min_viewed_t1,
  Age_t2, Min_t2,
  Rec_t2,
  Age_t3, Min_t3
)

head(data_3t_new)
```

```
##   Preference   Age_t1 Min_viewed_t1   Age_t2   Min_t2 Rec_t2   Age_t3   Min_t3
## 1 -0.6264538 17.86901      17.62655 18.86901 18.69849      1 19.86901 31.01163
## 2  0.1836433 35.03313      24.30205 36.03313 29.65043      1 37.03313 32.37940
## 3 -0.8356286 16.57445      15.20630 17.57445 15.03477      1 18.57445  7.87413
## 4  1.5952808 39.43825      31.11052 40.43825 37.40031      0 41.43825 44.44297
## 5  0.3295078 38.94124      28.71590 39.94124 29.87498      0 40.94124 31.43906
## 6 -0.8204684 20.09811      18.70748 21.09811 18.18643      1 22.09811 27.24766
```

```r
summary(data_3t_new)
```

```
##    Preference            Age_t1       Min_viewed_t1         Age_t2
##  Min.   :-3.671300   Min.   : 2.396   Min.   : 0.1989   Min.   : 3.396
##  1st Qu.:-0.669355   1st Qu.:24.598   1st Qu.:22.6351   1st Qu.:25.598
##  Median :-0.015493   Median :29.861   Median :27.3520   Median :30.861
##  Mean   :-0.003188   Mean   :29.921   Mean   :27.3555   Mean   :30.921
##  3rd Qu.: 0.697524   3rd Qu.:35.203   3rd Qu.:32.0894   3rd Qu.:36.203
##  Max.   : 3.810277   Max.   :58.995   Max.   :52.0274   Max.   :59.995
##      Min_t2           Rec_t2          Age_t3          Min_t3
##  Min.   :-4.324   Min.   :0.000   Min.   : 4.396   Min.   :-6.064
##  1st Qu.:23.800   1st Qu.:0.000   1st Qu.:26.598   1st Qu.:23.505
##  Median :29.881   Median :1.000   Median :31.861   Median :30.332
##  Mean   :29.916   Mean   :0.513   Mean   :31.921   Mean   :30.307
##  3rd Qu.:36.060   3rd Qu.:1.000   3rd Qu.:37.203   3rd Qu.:37.295
##  Max.   :63.696   Max.   :1.000   Max.   :60.995   Max.   :66.009
```

```r
write.csv(data_3t_new,
          file = "three_timepoint_simulation.csv",
          row.names = FALSE)
```